

Data Thinning

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Model Selection & Risk Minimization

- Risk:

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$$R_n(f) := \frac{1}{n} \sum_{i=1}^n \mathbb{E} [\ell_f(X_i, Y_i)] .$$

- Empirical Risk:

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$$\hat{R}_n(f) := \frac{1}{n} \sum_{i=1}^n \ell_f(X_i, Y_i)$$



- Problem:

- Same data →
 - Biased Estimator

- Solution:

- Data splitting

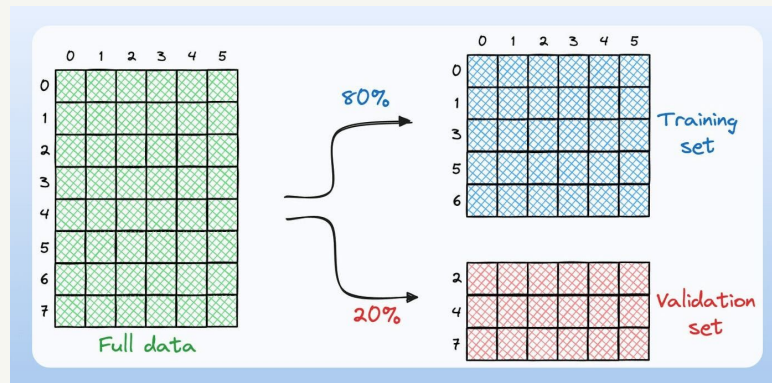
- Empirical Risk Minimizer:

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$$\hat{f} := \arg \min_{f \in \mathcal{F}} \hat{R}_n(f).$$

Data Splitting

- Standard solution
 - Train & Test datasets
- Advantage:
 - Independent datasets → unbiased estimator
- Limitations:
 - Reduces sample size
 - Perform poorly when sample sizes are small and large number of features
 - Non-separable dataset
 - Eg. spatial or time series data



Data Thinning

- Data Thinning constructs two independent datasets from one dataset by adding constructed noise instead of reducing sample size
- Suppose a dataset with n samples:
 - $(X_1, Y_1), \dots, (X_n, Y_n)$ where $X_i \in \mathbb{R}^p$ and $Y_i \in \mathbb{R}$,
 - We made assumption: normal distribution with unknown means and known variance
 - $Y_i \sim N(\mu_i, \sigma^2)$
 - generate noise
 - $W_i \sim N(0, \sigma^2)$
- Construct independent train and test datasets, positive γ
 - $Y_i^{(1)} = Y_i + \gamma W_i, \quad Y_i^{(2)} = Y_i - \frac{1}{\gamma} W_i.$

Research Question

1. How does data thinning affect the linear regression estimator?
2. How does data thinning affect the risk estimator?

Result 1: Unbiased Estimator

- Using thinned training data
 - $Y_i^{(1)} = Y_i + \gamma W_i$
- OLS estimator:
 - $\hat{\beta}_{DT} = (X^T X)^{-1} X^T Y^{(1)}$
- Data thinning estimator is an unbiased estimator
 - $E[\hat{\beta}_{DT}] = \beta^*$
 - Where $\beta^* = (X^T X)^{-1} X^T \mathbb{E}[Y]$.

Result 2: Inflated Variance

- Data thinning result in an unbiased estimator, but it increases the variance due to the noise added

- $$\text{Var}(\hat{\beta}_{DT}) = (1 + \gamma^2)\sigma^2(X^T X)^{-1}$$

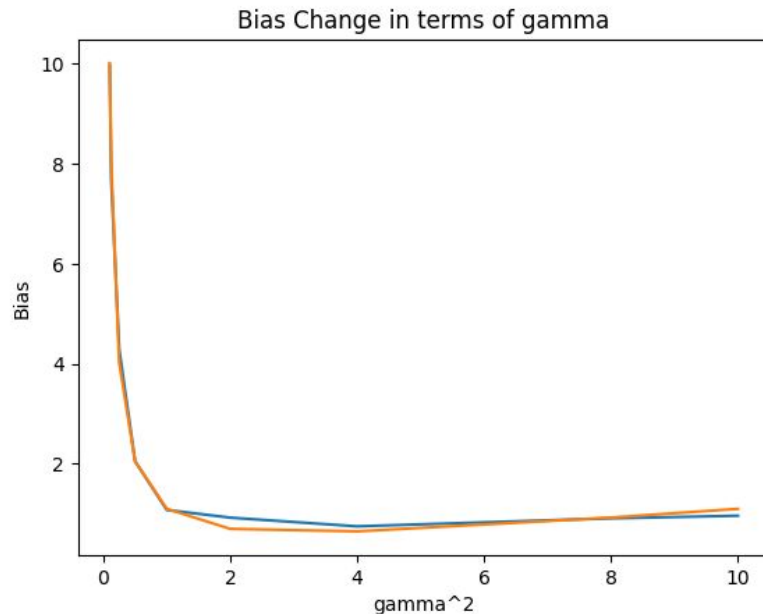
Result 3: Bias in Risk Estimation

- Goal: find the true risk
 - $R(\hat{\beta})$
- Reality: only able to obtain data thinning risk estimates
 - $\hat{R}^{(2)}(\hat{\beta}^{(1)})$
- Bias: difference between the true risk and the estimated risk

- $$\mathbb{E}_{W_1, \dots, W_n} \left[\hat{R}^{(2)} \left(\hat{\beta}^{(1)} \right) \right] - R \left(\hat{\beta} \right) = \frac{\sigma^2}{\gamma^2} + \frac{\sigma^2 \gamma^2 p}{n}$$
- $$= \underbrace{\left\{ \mathbb{E}_{W_1, \dots, W_n} \left[\hat{R}^{(2)} \left(\hat{\beta}^{(1)} \right) \right] - \mathbb{E}_{W_1, \dots, W_n} \left[R \left(\hat{\beta}^{(1)} \right) \right] \right\}}_{\text{Bias from noisy testing data}} + \underbrace{\left\{ \mathbb{E}_{W_1, \dots, W_n} \left[R \left(\hat{\beta}^{(1)} \right) \right] - R \left(\hat{\beta} \right) \right\}}_{\text{Bias from noisy training data}}$$
- n = sample size
- p = number of features

Simulation

- $n = 100$
 $d = 10$
 $X_{ij} \sim N(0, 1)$
 $\beta_i^* = 2^{-i+1}$
- Generate thinned datasets
- Data thinning OLS estimate using $Y^{(1)}$
- Estimate risk using $Y^{(2)}$
- we compute the empirical risk and true risk
 - *True risk is by drawing a new dataset
- Repeat for 1000 times



Takeaways

- Data thinning is an alternative approach of data splitting
- Under linear regression setting, the data thinning estimator $\hat{\beta}_{DT}$ remains unbiased, with increased variance
- Risk estimation is biased
 - $\frac{\sigma^2}{\gamma^2} + \frac{\sigma^2 \gamma^2 p}{n}$
- We hope to continue research on models beyond linear regression such as logistic regression, or other more complex models

Thank You!

Q&A